



New Site Mapping Deployment – Customer Operation Manual



深圳煜禾森科技有限公司

Shenzhen yuhesen Technology Co., Ltd

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1 Site Evaluation

Information	Note
Specific Site Location	On Google Maps, search for the desired location. Open the location details page, select Share , then copy the provided link. E.g. https://maps.app.goo.gl/AhFTpc8K7rvdfZZJ6
Navigation Map Inspection Area	
Narrow Passage	Narrow passages (less than 2 m). Mark the location of each passage and indicate the minimum width.
Slope	Mark areas with slopes (greater than 10 °) or provide supplementary photos.
Smooth or reflective surfaces (glass, mirrors, etc.)	None.
Featureless open areas: Large open spaces without obstacles (e.g., plazas) or environments with uniform structures and few distinct features (e.g., long undecorated corridors)	None.

2 Route Planning for Mapping

2.1 Preparations

Weather	Ensure no rain, snow, fog, or sandstorms, and that the ground surface is free of large water accumulation.
Power	Mapping duration depends on the site size and environmental complexity. Ensure both the robot and remote controller are fully charged.
Robot	Verify that the robot is operating properly. Check and clean the LiDAR sensor surface before use.

2.2 Personnel and Operation Guidelines

1. Non-essential personnel should stay outside the mapping area to prevent LiDAR interference, which may affect subsequent navigation.
2. When using the remote controller, avoid excessive speed and minimize in-place rotations. Use low-speed circular motion instead of sharp in-place rotations. Perform in-place rotation only when necessary in doorways.
3. It is recommended to have two operators: one controlling the robot with the remote controller, and one monitoring the mapping progress on the app. All personnel must remain within the robot's field of view from behind the robot during mapping.

2.3 Route Planning Standards

1. Mapping Initialization

When planning mapping routes in the app, wait until the **red LiDAR scan lines** appear. Then use the remote controller to guide the robot for mapping.



Figure 2.3.1

2. Slope Handling

During mapping, do not stop on slopes, as the LiDAR may scan the ground surface.



Figure 2.3.2

3. Charging Room Mapping Guidelines

a. Charging Room Setup

- i. **Entrance width:** The entrance must be at least 60 cm wider than the robot body. If there is a ramp at the entrance, the slope should be less than 10° to ensure smooth entry.
- ii. **Room size:** The charging room should be larger than 3 m × 4 m, with a depth of 4 m and width of 3 m. Leave sufficient turnaround space for the robot to adjust its position.
- iii. **Room height:** Ensure the LiDAR can scan normally without obstruction (LiDAR installation height recommended at 1 m or above).
- iv. **Room environment:** The charging room should have enough reflective features (e.g., walls, fixed objects). Avoid frequent movement of items inside the room. Room variation should be less than 10%. To prevent LiDAR misreading, do not use reflective transparent materials (e.g., glass) as walls. The floor should remain flat and even to reduce tilting or slipping, ensuring accurate positioning.
- v. **Charger location:** The charger must be securely fixed to the ground or wall to avoid shifting during docking. It is recommended to place it centrally against the entrance wall. When setting the charging point, leave about 1 m distance from the charger and ensure no obstacles nearby.
- vi. **Docking method:** When setting the charging point, use the remote control to manually align the robot with the charger. The robot's charging contacts must align with the charger contacts centrally and vertically.

b. Charging Room Mapping

Ensure the charging room door remains fully open during mapping, consistent with patrol conditions.

c. Charging Room Scanning

Requires two entries. For the first entry, drive the robot forward into the charging room until fully inside, then drive it backward out. For the second entry, drive the robot backward into the charging room, then drive it forward out.

4. Data Collection Completeness

During mapping, make sure that all required path data is collected for every target point. Avoid both missing data and unnecessary duplicate collection of the same route. **After all target points have been collected, ensure that additional routes needed to keep the map fully connected are also covered.**

- In the example below, the blue path represents the main route that passes through all target points. To maintain connectivity, the additional red path needs to be collected as well.
- If there are many side paths without target points, select a representative subset for collection. Choose evenly distributed paths, preferably those farther from the starting point. For example, in the figure only one of the three side branches was selected.

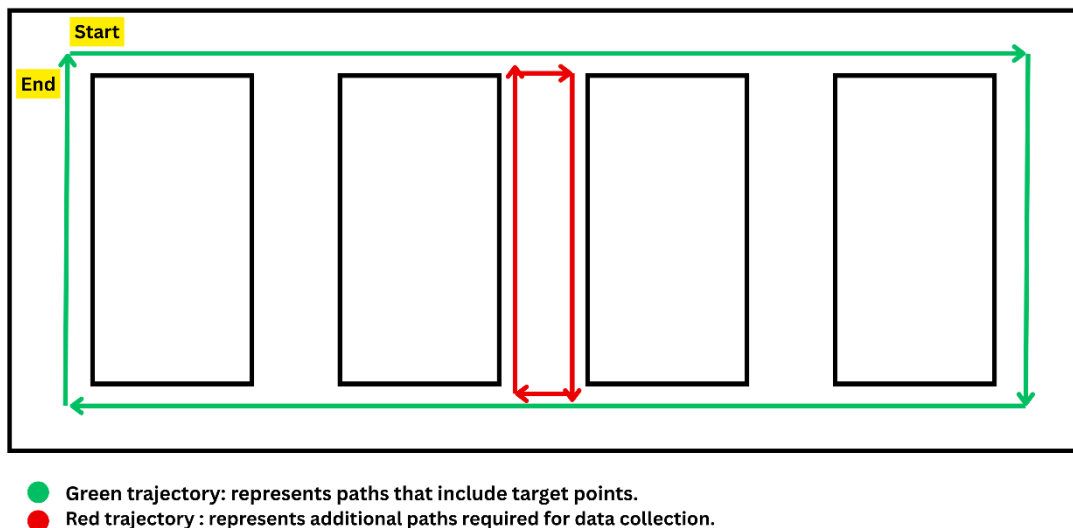


Figure 2.3.3

5. Mixed Indoor-Outdoor Environments

If the mapping area includes both indoor and outdoor sections (for example, covering an outdoor zone and a charging room together), set the mapping starting point outdoors.

6. Large Mapping Areas

For large mapping areas, first collect the **main outer loop** (perimeter framework), then proceed to collect interior details (inner paths and narrow passages).

Main Path Data Collection

When collecting data on the main path, drive the robot along the center of the path. After completing one full loop of the main path, continue driving forward for about 10 meters from the starting point along the same trajectory. This creates an effective overlap area, improving mapping accuracy.



Figure 2.3.4

Internal Path Data Collection

When collecting data on internal path, ensure that internal paths connect with the main path at as many points as possible. In these connection areas, the robot's position and heading should be consistent with its trajectory on the main path, forming reliable overlap regions.

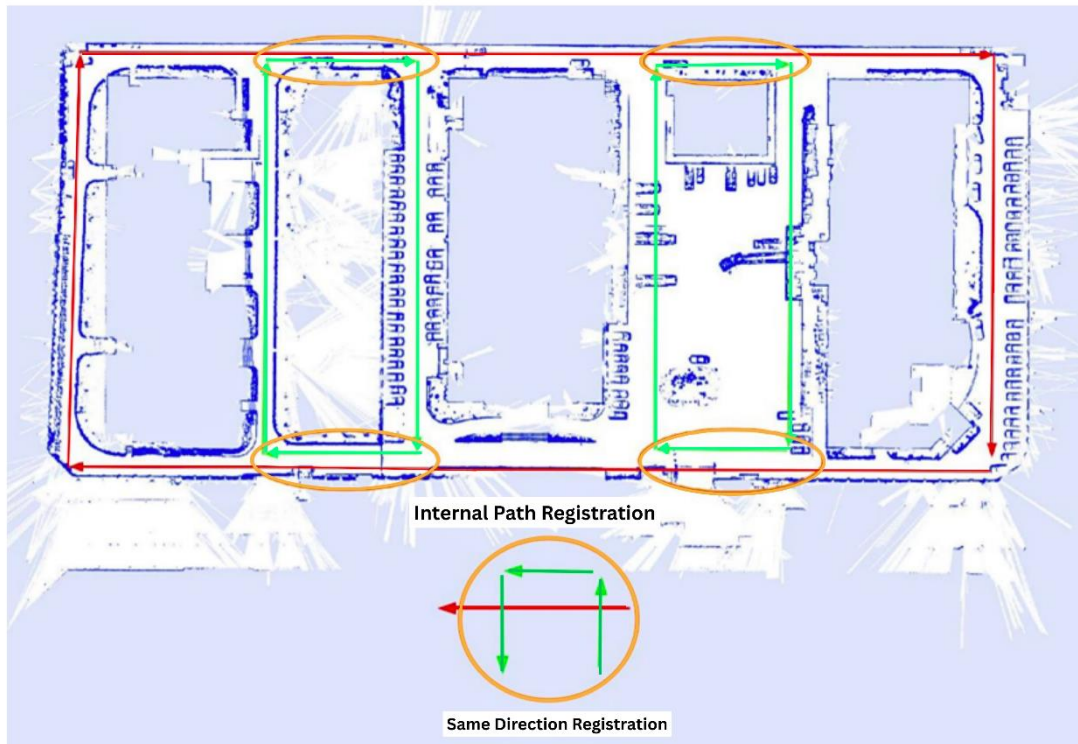


Figure 2.3.5

Dead-End Path Data Collection

If a dead-end path cannot directly connect with the main path, drive the robot as close as possible to the dead end. This ensures the dead-end path remains aligned and well integrated with the main path data.

7. Overlap at Trajectory Intersections

At intersections where different trajectories meet, ensure sufficient overlap to maintain mapping accuracy. Avoid trajectories that are either spaced too far apart or misaligned in heading. At intersections, the robot should move in a way that keeps orientation aligned, as shown in the figure below.

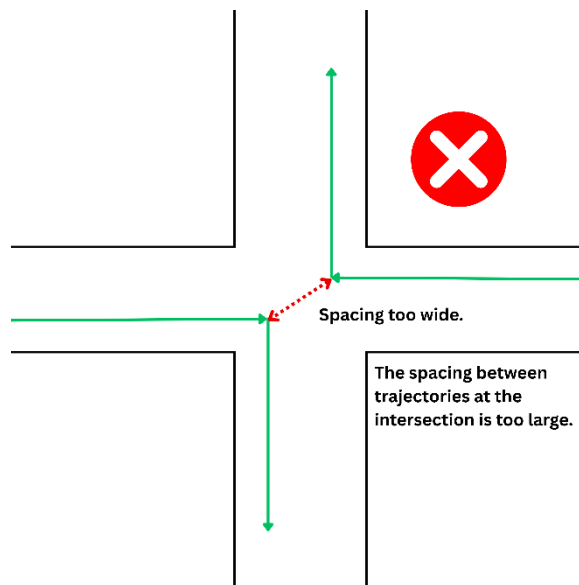


Figure 2.3.6

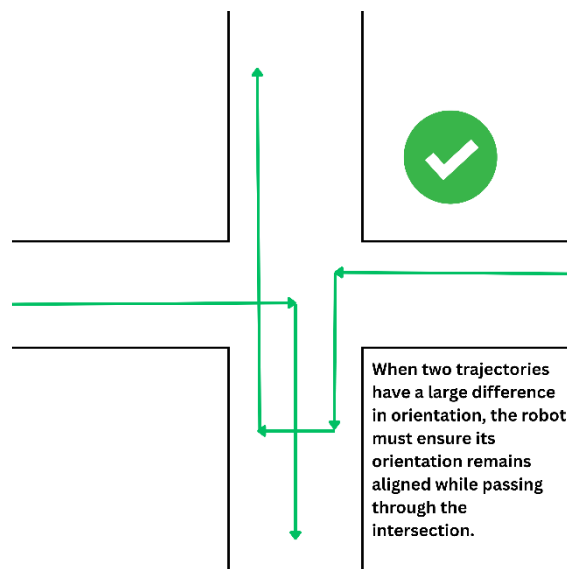
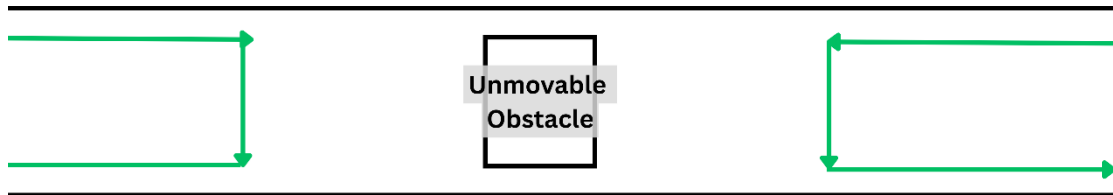


Figure 2.3.7

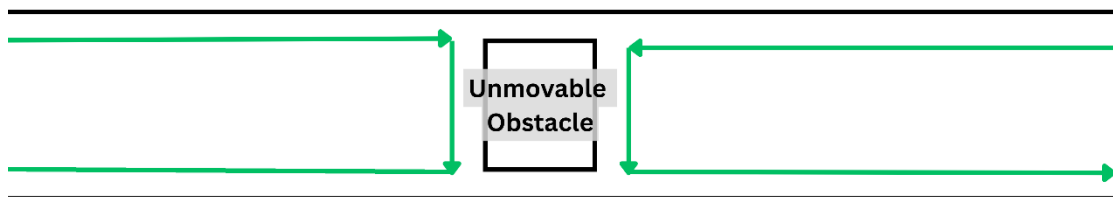
8. Unmovable Obstacles

When encountering unmovable obstacles, narrow passages, or dead-end paths that prevent trajectory continuity, manually drive the robot along paths as close as possible to these blocked areas.



If an unmovable obstacle is encountered, do not perform a turn from too far away. This results in trajectories near the obstacle having large orientation differences.

Figure 2.3.8



When encountering an unmovable obstacle, collect data by driving close to the obstacle. Ensure trajectories on both sides remain as close as possible to the obstacle, and maintain consistent robot orientation.

Figure 2.3.9

Sample Scenario

First collect the **main outer loop** (perimeter framework), then proceed to collect interior details (inner paths and narrow passages).

In the highlighted green area: when traversing a path for the second time, ensure it overlaps clearly with the first trajectory, including both the **same spatial position** and the **robot's orientation**. This overlap is essential for SLAM to maintain loop closure and pose graph consistency.

If the robot's heading is the same as in the first traversal, simply ensure an overlap region of at least 3 meters for reliable alignment (see figure highlighted in green).



Figure 2.3.10

If the robot's heading is opposite to the first traversal, manually drive the robot back and forth (with turn) once to create an overlap region of at least 3 meters with consistent heading. This helps maintain robust scan matching and map continuity.

2.4 Other Requirements and Guidelines

1. Do not place any objects on top of the robot, especially not on the LiDAR sensor.
2. When operating the robot manually, avoid in-place rotations and sharp U-turns. Try to make wider, smoother turns to keep the trajectory stable.
3. Personnel should not walk in front of or beside the robot during operation. It is recommended to follow behind the robot.
4. Non-essential personnel must stay away from the mapping area.
5. Since the installation height of the LiDAR may vary, some low obstacles such as steps or curbs may not be detected during mapping. After mapping, manually edit the map to add **Virtual Walls** at these locations to prevent the robot from driving onto them.
6. When operating indoors, if there are glass doors or glass walls, manually edit the map afterward to add **Virtual Walls** at these locations to prevent collisions.

3 Verification and Map Correction

3.1 Checking Map Quality During Mapping

The localization accuracy of the robot during navigation largely depends on the accuracy of the map. Therefore, building a high-quality map is essential. After the map has been generated along the planned mapping route, do not save it immediately. Instead, first check the quality of the generated map.

Key Checks

1. Whether the map shows obvious overlaps.
2. Whether walls display visible cracks or misalignments.

As shown in Figure 3.1, if cracks or misalignments appear on the walls after mapping, manually drive the robot back and forth along the affected wall to rescan. Once the correction is complete, the updated map should appear as in Figure 3.2.

3. If the map shows severe drift and cannot be corrected, it is recommended to rebuild the map.

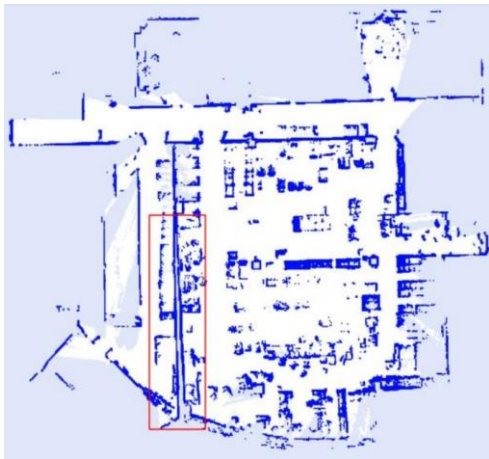


Figure 3.2.1



Figure 3.2.2

3.2 Post-Mapping Editing

1. Check if the mapped path contains noise points, doors, or other obstacles, and remove them promptly.

Editing Original Map

- Noise points appear in corridors where the robot has passed. These should be manually removed using the Editing original map function, as shown below.

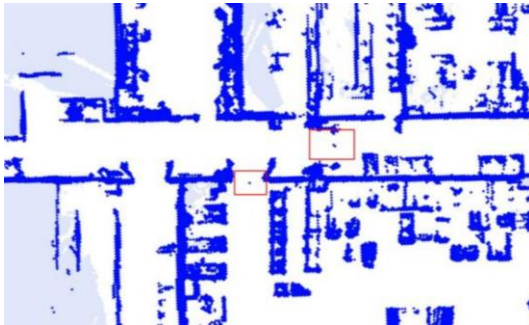


Figure 3.2.3



Figure 3.2.4

- Grey unscanned regions circled in red also need to be cleared to ensure a clean map. This can be done using the same function.

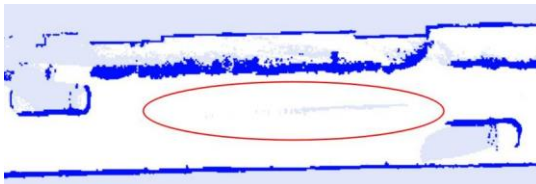


Figure 3.2.5

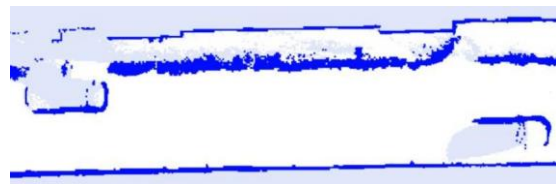
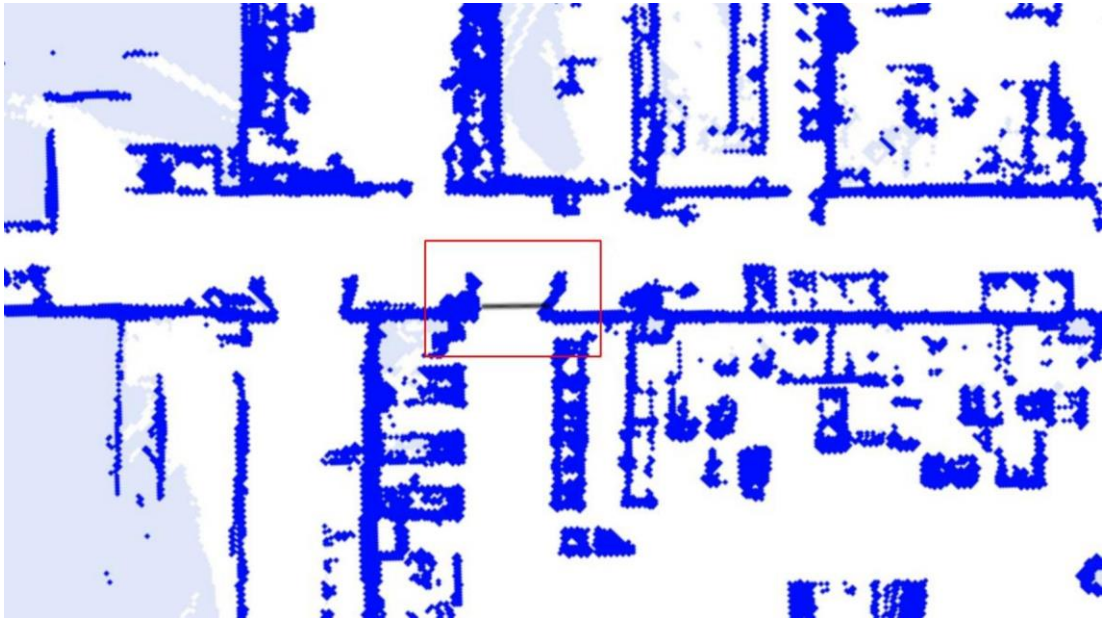


Figure 3.2.6

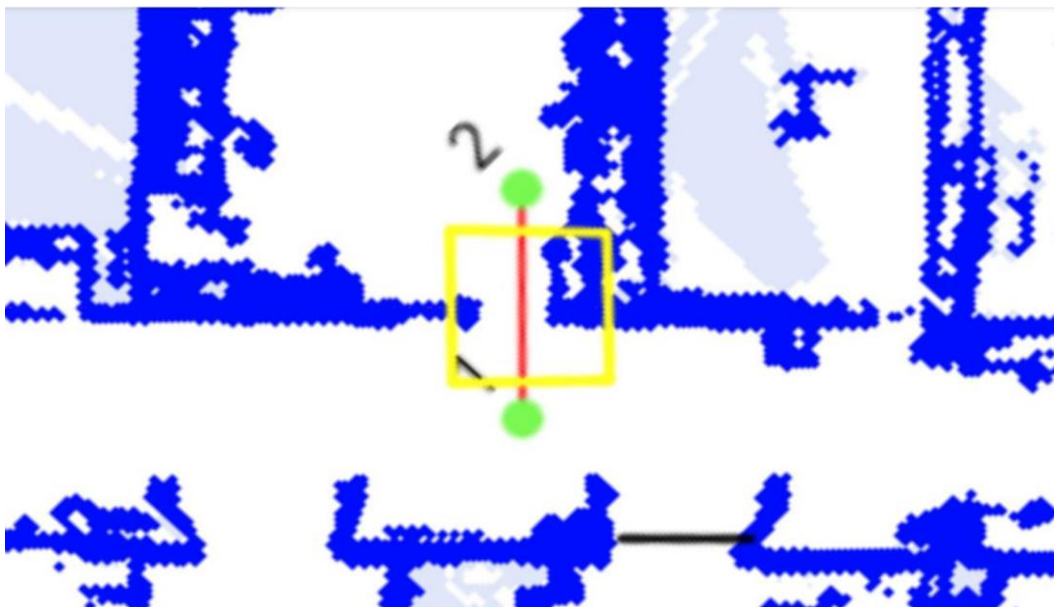
- During mapping, if people were accidentally scanned into the map, these can be removed in the same way to restore accuracy.
- When the map contains low obstacles that cannot be detected by the robot's sensors, or areas where the robot is not permitted to enter, virtual walls can be drawn.

*Figure 3.2.7*

2. Check if Narrow Passage and Slope areas are mapped correctly

Narrow Passages

When the robot needs to pass through a doorway, a doorway area can be defined as shown in Figure 1 (yellow boxed region). The defined area should be drawn slightly larger than the actual doorway size to ensure smooth navigation. When robot is navigating through doorways, it is recommended to use the Draw Path from Path and Point Management. A manually drawn path should be placed near the center of the doorway to guide the robot, see the red line in Figure 1. The minimum doorway clearance shall be at least 400 mm greater than the robot body width.

*Figure 3.2.8*

Long Straight Corridor

As shown in Figure 2, when the robot needs to pass through an extended straight passage (approximately 5 m in length), a corridor region should be drawn. The marked area should be slightly larger than the actual corridor to ensure reliable path planning. When navigating such a passage, it is recommended to use manual path drawing mode. As illustrated, draw a path along the center of the corridor to maintain alignment. The minimum clearance shall be at least 400 mm greater than the robot body width.

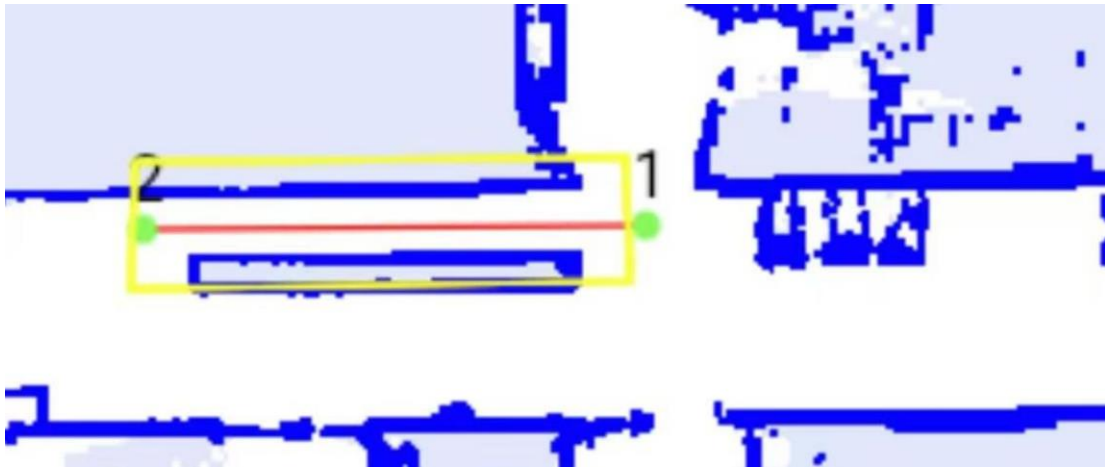


Figure 3.2.9

Note

- If the corridor length exceeds 50 m, the robot may lose localization. This is due to the corridor's uniform structure and the limited effective sensing range of LiDAR (within approximately 50 m). As the distance increases, the resolution of the LiDAR data decreases.

To improve localization, additional methods may be applied, such as adding distinct visual markers or introducing identifiable features within the long corridor to assist the robot in maintaining accurate positioning.

Slope Area Setup

As shown in Figure 2, when the robot needs to traverse a slope greater than 5° , a slope area must be drawn. The drawn area should be slightly larger than the actual slope to ensure full coverage of the real slope surface.

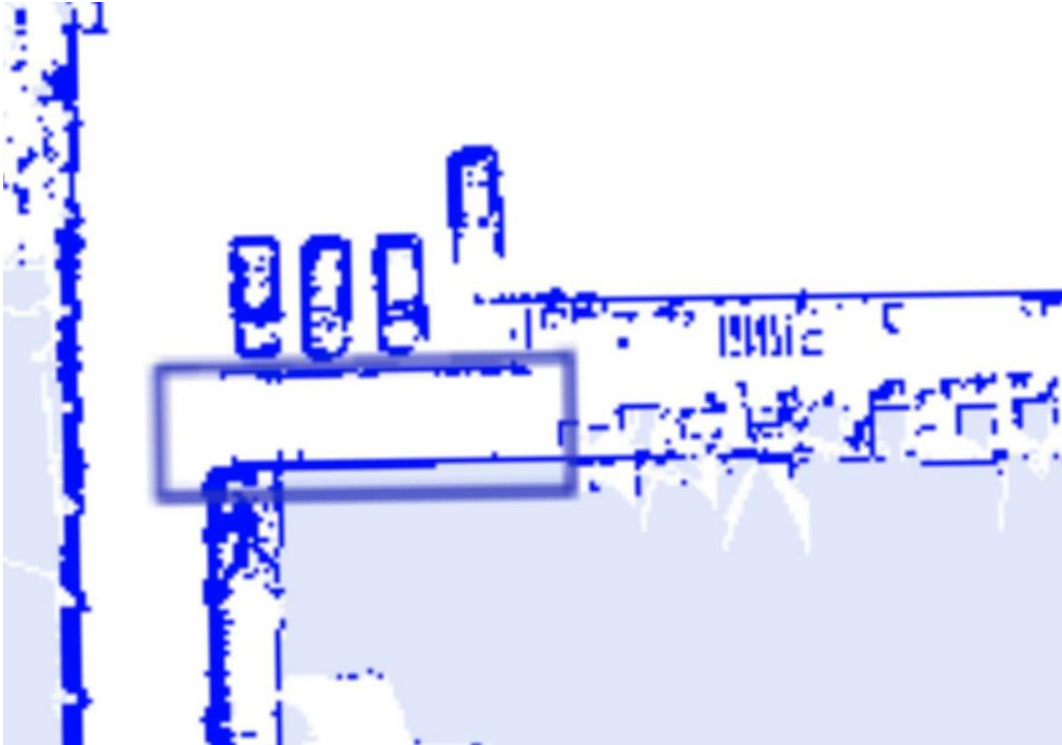


Figure 3.2.10